

Symmetry premiums, cyclic-closure coverage, and defect-spectrum persistence in Kautz digraphs

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Abstract

Let an involution of type $2^m 1^f$, $m \geq 1$, act on length-three Kautz words by letterwise involution followed by reversal. We prove that the minimum invariant feedback vertex number is the ordinary value plus m^2 , with surcharge $m + m(m - 1)$ split into local orbit forcing and a global acyclicity cost. Every invariant optimum is the full cyclic closure of an ordinary optimum exactly when $m = 1$ in the nonidentity domain; the identity case is stated separately. For the order-cone defect spectrum we prove the monotonicity

$$\mathcal{D}_{r,0} \subseteq \mathcal{D}_{m,f} \quad (m \geq r \geq 1, f \geq 0).$$

An exact finite census at $(m, f) = (3, 0)$, accompanied here by one explicit certificate for each value, therefore implies that defects 0, 1, 2, 3, 4 are all attained for every $m \geq 3$ and $f \geq 0$. This last conclusion is attainability only: no equality of later spectra, general upper bound four, or interval theorem is asserted. More generally, defect is the transversal number of the clutter of minimal infeasible bad-cell constraints. Forced cells and a matching of pairwise incompatibilities give a sound lower certificate, while a four-letter rank-three example shows that this certificate need not be exact. At the six-letter base, a ten-relation precedence core forces four bad triangle cells by a $2 + 1 + 1$ argument and occurs in exactly 24 of the 1,759 symmetry classes in the census.

Keywords: Kautz digraphs; feedback vertex sets; reverse-complement symmetry; cyclic closure; defect spectra; exact census.

MSC 2020: 05C20 (primary); 05C69 (secondary).

Reproducibility package: <https://github.com/aconsciousfractal/FCIG-Symmetry-Premiums-Cyclic-Closure-Coverage-and-Defect-Spectrum-Persistence-in-Kautz-Digraphs>

1 Introduction

Feedback vertex sets in Kautz digraphs admit a useful line-digraph description: a vertex deletion problem on length-three words becomes a feedback arc problem on length-two words. We study this problem under the reverse-complement symmetry obtained by reversing a word and applying a letter involution. Three questions then separate naturally: what is the exact cost of invariance, when are all invariant optima generated from ordinary optima by cyclic closure, and which losses occur inside the topological-order cone of an invariant optimum? Here the cyclic closure of a set F means $\text{cl}_\Theta(F) = F \cup \Theta_\sigma(F)$.

Our main contributions are as follows.

1. For an involution of type $2^m 1^f$, the invariant optimum exceeds the ordinary feedback vertex number by exactly m^2 . The surcharge splits as $m + m(m - 1)$ into a local orbit-forcing term and a global acyclicity term.

2. Complete cyclic-closure coverage holds exactly for one moving transposition block; when $m \geq 2$, closure-generated and non-closure-generated invariant optima coexist.
3. The defect spectrum is monotone under adjoining moving blocks and fixed letters. A certified six-letter base therefore yields persistent attained defects 0, 1, 2, 3, 4 for every $m \geq 3$.
4. Defect equals the transversal number of an exact obstruction clutter. Its singleton and pair members yield a forced-cell-plus-matching lower bound, which a rank-three four-letter obstruction shows need not always be tight.
5. A ten-relation order core gives a short human explanation for a family of extremal defect-four objects in the six-letter census.

The ordinary value and cycle-cell substrate are known [8, 9]. The feedback-arc/order and line-digraph correspondences are classical [11, 2]. We use only the elementary antisymmetrization device behind Section 9 of Goldberg–Karzanov [4]; no Kautz-specific theorem is imported from that paper. Adjacent work on symmetric decycling sets in binary de Bruijn graphs uses reverse complementation to define the k -nonical space [6]. Its graph family, alphabet, object and optimization target differ from ours; we import no result from it and infer no priority from that comparison. Earlier work by the present author proves an existence-and-parity criterion for literal reverse-complement-invariant ordinary minimum decycling sets in q -ary de Bruijn graphs [1]. The present paper instead treats length-three Kautz digraphs, allows an invariance premium over the ordinary optimum, and studies closure coverage and order-cone defect; no theorem is transferred between the two graph families. The full texts of two potentially relevant Kautz sources [10, 12] could not be obtained. We therefore make no claim of bibliographic priority or source completeness.

2 Ordinary Kautz substrate and notation

Let A be an alphabet of size $q \geq 2$. The vertices of $K(q-1, k)$ are the words $x_1 \cdots x_k$ with adjacent letters distinct, and an arc shifts one position to the left. Put

$$H = K(q-1, 2), \quad K(q-1, 3) = L(H).$$

Thus a word xyz is also the predecessor arc $xy \rightarrow yz$ of H . Write

$$S_{q-1} = \frac{q(q-1)(2q-1)}{6}, \quad \beta_q = \binom{q}{2} + 4\binom{q}{3}.$$

Proposition 2.1 (Ordinary substrate; known background). *The minimum feedback vertex number of $K(q-1, 3)$ is S_{q-1} , and the maximum number of forward predecessor arcs in a total order of $V(H)$ is β_q .*

Proof. The arcs of H partition into one directed digon for each unordered pair of letters and two oriented directed triangles for each unordered triple. Every feedback arc set meets every cell, giving $\binom{q}{2} + 2\binom{q}{3} = S_{q-1}$. Given a total order on A , delete the strict middle peaks xyz with $y > x, y > z$. Their number is $\sum_{i=0}^{q-1} i^2 = S_{q-1}$, and the standard descending-then-ascending order of the pairs xy makes the complement acyclic. Finally

$$q(q-1)^2 - S_{q-1} = \binom{q}{2} + 4\binom{q}{3} = \beta_q.$$

The result and construction are cited background; the proof fixes our normalization and the line-digraph crosswalk. $\square$

3 Symmetry, order cones, and defect

Let σ be an involution of A , of cycle type $2^m 1^f$, and put

$$\Theta_\sigma(xyz) = \sigma(z)\sigma(y)\sigma(x), \quad j(xy) = \sigma(y)\sigma(x).$$

On predecessor arcs the induced anti-action is $\rho(u, v) = (j(v), j(u))$. Let τ_σ be the minimum size of a Θ_σ -stable feedback vertex set and $\Delta_\sigma = \tau_\sigma - S_{q-1}$.

Let $\mathcal{C}$ denote the digon-and-oriented-triangle partition above and let $G_\sigma = \langle \Theta_\sigma \rangle$ act on its cells. For one representative C of each cell orbit, put

$$H_C = \text{Stab}_{G_\sigma}(C), \quad s(C) = \min_{v \in C} |H_C v|.$$

Proposition 3.1 (Invariant cell-transversal formula). *The minimum size of an invariant cell transversal is*

$$L_\sigma = \sum_{[C] \in \mathcal{C}/G_\sigma} |G_\sigma C| s(C).$$

Consequently

$$\ell_\sigma = L_\sigma - S_{q-1}, \quad g_\sigma = \tau_\sigma - L_\sigma, \quad \Delta_\sigma = \ell_\sigma + g_\sigma.$$

Proof. Let X be an invariant cell transversal and take $v \in X \cap C$. Invariance forces $G_\sigma v \subseteq X$, while the cell partition gives $\text{Stab}_{G_\sigma}(v) \leq H_C$. Hence orbit-stabilizer gives

$$|G_\sigma v| = [G_\sigma : H_C] [H_C : \text{Stab}_{G_\sigma}(v)] = |G_\sigma C| |H_C v|.$$

The inverse images of distinct cell orbits are disjoint, so summing gives the lower bound. Conversely, choose in every representative cell a vertex attaining $s(C)$ and take the union of its full G_σ -orbit. This invariant set meets every cell and attains the displayed sum. Finally, the cells are disjoint and their number is S_{q-1} , whereas every feedback set meets every cell. Thus $S_{q-1} \leq L_\sigma \leq \tau_\sigma$, proving the decomposition and nonnegativity. $\square$

Let P be a maximum ρ -stable acyclic predecessor arc set. For $\omega \in \text{Top}(P)$, write A_ω for all ambient arcs of H directed forward by ω , and define

$$\delta_H(P) = \beta_q - \max_{\omega \in \text{Top}(P)} |A_\omega|. \quad (3.1)$$

Let $b(\omega)$ count the oriented-triangle cells having only one forward arc in ω .

Lemma 3.2 (Bad-triangle formula). *For every total order ω , $|A_\omega| = \beta_q - b(\omega)$; hence $\delta_H(P) = \min_{\omega \in \text{Top}(P)} b(\omega)$.*

Proof. Every digon has one forward arc. Every directed triangle has one or two; if b of the $2\binom{q}{3}$ triangles have one, the score is $\binom{q}{2} + b + 2(2\binom{q}{3} - b) = \beta_q - b$. $\square$

For an ordinary feedback set F , put $\text{cl}_\Theta(F) = F \cup \Theta_\sigma(F)$.

Lemma 3.3 (Order-cone dictionary; standard-derived). *The invariant minimum complementary to P is a full cyclic closure of an ordinary minimum if and only if $\delta_H(P) = 0$.*

Proof. Every acyclic arc set lies in the forward set of a topological order, and it lies in an ordinary maximum precisely when such an order has score β_q . If $P \subseteq A_\omega$ with that score, stability gives $P \subseteq A_\omega \cap \rho(A_\omega)$; the intersection is stable and acyclic, so maximality forces equality. Complementation gives the full closure of the ordinary minimum $E(H) \setminus A_\omega$. Conversely, if the complement of P is such a closure, then P is contained in the complementary ordinary maximum and therefore has a topological order of score β_q . $\square$

4 Exact symmetry premium and its split

For the fixed-point-free case write the transposition blocks as $B_i = \{a_i, b_i\}$, $0 \leq i < m$, and $\bar{a}_i = b_i$.

Theorem 4.1 (Exact premium and local/global split). *If σ has type 2^m , $m \geq 1$, on $q = 2m$ letters, then*

$$\tau_\sigma(K(2m-1, 3)) = S_{2m-1} + m^2, \quad \Delta_\sigma = m^2, \quad \ell_\sigma = m, \quad g_\sigma = m(m-1).$$

Proof. By Lemma 7.1 below, a maximum stable predecessor DAG is $I(h) = \{u \rightarrow v : h(u) < h(v)\}$ for an antisymmetric potential $h \circ j = -h$. The fixed predecessor vertices $a_i b_i, b_i a_i$ have level zero. Classify an arc $xy \rightarrow yz$ by the set of transposition blocks meeting $\{x, y, z\}$. Its block support has size one, two, or three, and the following table is an exhaustive disjoint partition. The counts and bounds are for each fixed choice of block support.

support	pieces	number	bound per piece
one block	within-block digon	1	0
two blocks	fixed-free two-arc pairs	8	1
	cross-block digons	4	1
	non-digon directed four-cycles	2	3
three blocks	directed triangles	16	2

Indeed, in one block both predecessor vertices are fixed and tied at zero. For two blocks there are eight nonfixed predecessor vertices; each supports one incoming arc from a fixed vertex and one outgoing arc to a fixed vertex, so at most one of the two can increase. The remaining arcs are exactly four digons and two directed four-cycles. For three blocks they are exactly the two oriented triangles on each of the eight choices of one letter per block. An increasing potential selects at most one arc of a digon, three of a directed four-cycle, and two of a directed triangle. Therefore

$$|I(h)| \leq B_m := 18 \binom{m}{2} + 32 \binom{m}{3} = 4m(m-1) + 10 \binom{m}{2} + 32 \binom{m}{3} = \frac{m(m-1)(16m-5)}{3}. \quad (4.1)$$

This bound is sharp. Put $\lambda_i = 4^{m-1-i}$, set $c_{ij} = 2\lambda_j$ for $i < j$, $c_{ji} = -c_{ij}$, and identify $i^- = a_i, i^+ = b_i$. For $i \neq j$ define

$$h(i^\varepsilon j^\delta) = \varepsilon \lambda_i + \delta \lambda_j + c_{ij} \quad (\varepsilon, \delta \in \{-1, +1\}). \quad (4.2)$$

Complete the definition by $h(i^- i^+) = h(i^+ i^-) = 0$. Since $j(i^\varepsilon j^\delta) = j^{-\delta} i^{-\varepsilon}$ and $c_{ji} = -c_{ij}$, this is antisymmetric. If $i < j$, the sign of $h(i^\varepsilon j^\delta)$ is ε , since $\lambda_i \geq 4\lambda_j$. Thus all eight fixed-free pairs in that two-block sector attain their bound.

Fix $i < j$ and let $\varepsilon, \delta, \eta, \zeta \in \{-1, +1\}$. On the two alternating arc families

$$i^\varepsilon j^\delta \rightarrow j^\delta i^\eta, \quad j^\delta i^\varepsilon \rightarrow i^\varepsilon j^\zeta,$$

the potential differences are respectively

$$(\eta - \varepsilon)\lambda_i - 4\lambda_j, \quad (\zeta - \delta)\lambda_j + 4\lambda_j;$$

the first is positive exactly for $(\varepsilon, \eta) = (-1, +1)$, giving two arcs as δ varies, while the second belongs to $\{2\lambda_j, 4\lambda_j, 6\lambda_j\}$ and is positive for all eight choices. Hence the four digons and two four-cycles contribute exactly $2 + 8 = 10$.

Finally fix $i < j < k$ and $\varepsilon, \delta, \gamma \in \{-1, +1\}$. Around the directed triangle

$$i^\varepsilon j^\delta \rightarrow j^\delta k^\gamma \rightarrow k^\gamma i^\varepsilon \rightarrow i^\varepsilon j^\delta$$

the three potential differences are

$$-\varepsilon\lambda_i - 2\lambda_j + (\gamma + 2)\lambda_k, \quad \varepsilon\lambda_i - \delta\lambda_j - 4\lambda_k, \quad (\delta + 2)\lambda_j + (2 - \gamma)\lambda_k.$$

Their signs are $(-\varepsilon, \varepsilon, +)$, since $\lambda_i \geq 4\lambda_j \geq 16\lambda_k$; exactly two arcs increase in each of these eight triangles. Antisymmetry makes $I(h)$ ρ -stable; the other eight oriented triangles are their ρ -images as the signs vary, and have the same count. Thus every row of the exhaustive sector table is attained.

Since H has $2m(2m - 1)^2$ arcs, complementation gives

$$\tau_\sigma = 2m(2m - 1)^2 - B_m = S_{2m-1} + m^2.$$

For the local term, precisely the m within-block digon cells are fixed with their two vertices exchanged, forcing two rather than one selected vertex. Every other cell orbit has trivial internal stabilizer and attains the ordinary cell cost. Proposition 3.1 gives $L_\sigma = S_{2m-1} + m$, and hence $\ell_\sigma = m$ and $g_\sigma = m^2 - m$. $\square$

5 Fixed letters and zero/one constructions

Lemma 5.1 (Exact fixed-letter transport). *Adjoining one σ -fixed letter t to an alphabet B of size n increases the ordinary and invariant feedback optima by n^2 , preserves the symmetry premium and its local/global split, and sends a canonical lifted maximum predecessor DAG to one of exactly the same defect.*

Proof. For a feedback set M_B , put

$$M_{B+t} = M_B \cup X_t, \quad X_t = \{xtz : x, z \in B\}. \quad (5.1)$$

Extend a topological order of the old residual by placing all ta before all old pairs and all at after them. The retained new words are forward, so (5.1) is a feedback set. The n new digons and $2\binom{n}{2}$ new oriented triangles are disjoint and total n^2 ; every feedback set must meet all of them. This proves the ordinary increment. The set X_t is invariant, and restriction of an invariant feedback set is invariant, giving the same invariant increment. Its unique middle- t choice meets each new cell orbit minimally, so the split is preserved.

Let $P^\uparrow$ be the residual of (5.1). Restricting a topological order of $P^\uparrow$ to B shows $\delta_{B+t}(P^\uparrow) \geq \delta_B(P)$, because new cells contribute at most $\beta_{n+1} - \beta_n$. The displayed extended order attains the maximum in every new cell, giving the reverse inequality. Thus the defect is preserved, and the construction iterates. $\square$

Corollary 5.2 (Premium with fixed letters). *For type $2^m 1^f$, $m \geq 1, f \geq 0, q = 2m + f$,*

$$\tau_\sigma(K(q - 1, 3)) = S_{q-1} + m^2, \quad (\ell_\sigma, g_\sigma) = (m, m(m - 1)).$$

Proof. Iterate Lemma 5.1 from Theorem 4.1. $\square$

Proposition 5.3 (Four-letter seeds). *For $A = \{0, 1, 2, 3\}$ and $\sigma = (01)(23)$, maximum invariant predecessor DAGs of defects zero and one both exist.*

Proof. Here $\beta_4 = 22$ and every invariant minimum has size 18. For defect zero, take the ordinary minimum

$$M = \{010, 012, 132, 202, 210, 212, 232, 301, 302, 303, 310, 312, 313, 320\}$$

and its closure

$$N_0 = M \cup \Theta(M) = \{010, 012, 101, 102, 103, 132, 202, 210, 212, 232, 301, 302, 303, 310, 312, 313, 320, 323\}.$$

The complement of M is forward in $10 < 02 < 12 < 20 < 01 < 03 < 32 < 21 < 13 < 23 < 30 < 31$, while the complement of N_0 is forward in $02 < 10 < 12 < 20 < 01 < 03 < 23 < 32 < 21 < 13 < 30 < 31$. Hence both are optimal and the residual of N_0 has defect zero by Lemma 3.3.

For defect one, let

$$N_1 = \{010, 013, 021, 030, 031, 032, 101, 102, 121, 123, 201, 202, 230, 232, 310, 313, 321, 323\}.$$

It is invariant, and its complement is forward in

$$21 < 10 < 13 < 30 < 01 < 02 < 23 < 31 < 12 < 32 < 20 < 03.$$

This order scores 21, with unique bad cell $\{031, 310, 103\}$. Defect zero would force an ordinary minimum $M \subseteq N_1$ whose cyclic closure is N_1 . Such an M contains exactly one word from each of the fourteen cycle cells. Ten singleton intersections force

$$202, 030, 121, 313, 201, 013, 230, 032, 123, 321.$$

The remaining digons force one of $010, 101$ and one of $232, 323$. The final two triangle choices must be either $\{021, 310\}$ or $\{102, 031\}$. In the first case the residual contains

$$312 \rightarrow 120 \rightarrow 203 \rightarrow 031 \rightarrow 312,$$

and in the second it contains

$$213 \rightarrow 130 \rightarrow 302 \rightarrow 021 \rightarrow 213.$$

Neither candidate is a feedback vertex set, independently of the two digon choices. Thus the defect is positive and therefore exactly one. $\square$

Theorem 5.4 (Uniform zero/one construction). *For every $m \geq 1, f \geq 0$, defect zero is attained. For every $m \geq 2, f \geq 0$, defect one is attained.*

Proof. For $m = 1$, the empty moved-core is the unique maximum and has defect zero. Lemma 7.1 and Proposition 7.2 represent this seed and both four-letter seeds of Proposition 5.3 as $I(h)$ for sharp antisymmetric potentials. Lemmas 7.3 and 7.4 below extend a sharp core by one moving block while preserving its exact defect. They therefore propagate the zero seed from $m = 1$, and the defect-one seed of Proposition 5.3 from $m = 2$, to every larger m . Lemma 5.1 then propagates both families over all f . The cited lemmas are independently proved in Section 7, so this forward reference is not circular. $\square$

6 Fixed-digon ceiling and closure coverage

Proposition 6.1 (Fixed-digon defect ceiling). *Every maximum ρ -stable predecessor DAG P of type $2^m 1^f$, $m \geq 1$, satisfies*

$$0 \leq \delta_H(P) \leq m(m-1).$$

Proof. For each moving block, the vertices $a_i b_i, b_i a_i$ are fixed by j , while the two arcs between them are exchanged by ρ . A stable DAG contains neither. Every topological order recovers exactly one forward arc from each such omitted digon. By Corollary 5.2, $|P| = \beta_q - m^2$, so every $\omega \in \text{Top}(P)$ has $|A_\omega| \geq \beta_q - m^2 + m$. Subtracting the maximum score from β_q proves the bound. $\square$

Theorem 6.2 (Cyclic-closure coverage). *For every nonidentity involution of type $2^m 1^f$, $m \geq 1, f \geq 0$: every maximum invariant predecessor DAG has defect in $[0, m(m-1)]$; a zero-defect maximum exists; an exact defect-one maximum exists for $m \geq 2$; and every invariant minimum is a full cyclic closure of an ordinary minimum if and only if $m = 1$.*

Proof. The bound is Proposition 6.1, and the two existence assertions are Theorem 5.4. If $m = 1$, the bound forces every maximum to have defect zero, and Lemma 3.3 gives complete coverage. If $m \geq 2$, the exact defect-one construction supplies an invariant minimum that is not closure-generated. The simultaneous zero-defect construction shows that existence of a closure-generated optimum and failure of complete coverage coexist. $\square$

Proposition 6.3 (Identity boundary). *If $\sigma = \text{id}$, $q = f \geq 2$, then $\tau_\sigma = S_{q-1}$, the surcharge is zero, and every invariant minimum is the full cyclic closure of an ordinary minimum.*

Proof. The strict middle-peak ordinary minimum from Proposition 2.1 is invariant under reversal, so the ordinary and invariant values agree. Any invariant minimum is itself an ordinary minimum and equals its own cyclic closure. $\square$

7 Sharp potentials and support extension

For a potential $h : V(H) \rightarrow \mathbb{R}$, write $I(h) = \{u \rightarrow v \in E(H) : h(u) < h(v)\}$.

Lemma 7.1 (Antisymmetric potential completion). *Let a finite digraph have a vertex involution j , with arc anti-action $\rho(u, v) = (j(v), j(u))$. If P is a ρ -stable acyclic arc set, then an antisymmetric potential $h \circ j = -h$ satisfies $P \subseteq I(h)$. If P has maximum cardinality among stable acyclic arc sets, then $P = I(h)$.*

Proof. Choose a strictly increasing topological rank t of P and put $h(v) = t(v) - t(j(v))$. Then $h(j(v)) = -h(v)$. If $u \rightarrow v \in P$, stability also gives $j(v) \rightarrow j(u) \in P$, and

$$h(v) - h(u) = [t(v) - t(u)] + [t(j(u)) - t(j(v))] > 0.$$

Thus $P \subseteq I(h)$. The set $I(h)$ is acyclic because h strictly increases on it, and it is stable by antisymmetry. Maximal cardinality now forces equality. $\square$

Proposition 7.2 (Sharp-core localization). *In fixed-point-free type 2^m , every maximum stable predecessor DAG is $I(h)$ for a sharp antisymmetric potential attaining B_m . Its restriction to any subset of moving blocks is maximum. Moreover, if P_{ij} is its two-block restriction, then*

$$\delta_H(P) \geq \sum_{i < j} \delta(P_{ij}). \quad (7.1)$$

Proof. Lemma 7.1 gives $P = I(h)$. Theorem 4.1 and (4.1) give $|P| = B_m$. Since (4.1) is the sum of disjoint sector bounds, equality forces saturation of every sector; summing the sectors supported on any chosen block subset gives its exact smaller bound. Thus every restriction is maximum. Finally, by Lemma 3.2, defect counts bad oriented triangles. The bad triangle cells supported on different unordered block pairs are disjoint (shared within-block digons have zero defect), and every global topological order restricts to one of each P_{ij} . Summing their local lower bounds proves (7.1). Equality in (7.1) is not asserted. $\square$

Lemma 7.3 (Sharp dominant moving-block extension). *Let h be an antisymmetric potential on $r \geq 1$ moving blocks with $|I(h)| = B_r$. Put $H_0 = \max_v |h(v)|$, $L = H_0 + 4$, and adjoin the block $\{a, b\}$. If $\varepsilon_x \in \{-1, +1\}$ is the sign of an old letter in its block, preserve all old levels and set*

$$\begin{aligned} \tilde{h}(ax) &= -L + \varepsilon_x + 2, & \tilde{h}(xa) &= -L + \varepsilon_x - 2, \\ \tilde{h}(bx) &= L + \varepsilon_x + 2, & \tilde{h}(xb) &= L + \varepsilon_x - 2, \\ \tilde{h}(ab) &= 0, & \tilde{h}(ba) &= 0. \end{aligned} \tag{7.2}$$

Then $\tilde{h}$ is antisymmetric and $|I(\tilde{h})| = B_{r+1}$.

Proof. Write the new letters as $n^- = a, n^+ = b$, and write an old letter as x^δ , where $\delta \in \{-1, +1\}$ records its sign in its moving block. Then (7.2) is equivalently

$$\tilde{h}(n^\varepsilon x^\delta) = \varepsilon L + \delta + 2, \quad \tilde{h}(x^\delta n^\varepsilon) = \varepsilon L + \delta - 2.$$

These formulas and $j(n^\varepsilon x^\delta) = x^{-\delta} n^{-\varepsilon}$ prove antisymmetry directly. Every new nonfixed level has absolute value at least $H_0 + 1$, so each of the eight fixed-free two-arc pairs belonging to a new/old block pair contributes exactly one arc.

For $\varepsilon, \zeta, \delta, \gamma \in \{-1, +1\}$, the two alternating families have differences

$$(\zeta - \varepsilon)L - 4, \quad \gamma - \delta + 4 \in \{2, 4, 6\};$$

the first is positive exactly when $(\varepsilon, \zeta) = (-1, +1)$, giving two arcs as δ varies, and the second is positive for all eight choices. Hence the alternating part attains ten, just as in the two-block row of the sector table.

Now choose two distinct old letters x^δ, y^γ from different old blocks. On the directed triangle $n^\varepsilon x^\delta \rightarrow x^\delta y^\gamma \rightarrow y^\gamma n^\varepsilon \rightarrow n^\varepsilon x^\delta$, its cyclic levels have the form

$$p = \varepsilon L + \delta + 2, \quad q \in [-H_0, H_0], \quad r = \varepsilon L + \gamma - 2.$$

If $\varepsilon = -1$, then $r < p < q$; if $\varepsilon = +1$, then $q < r < p$. Exactly two cyclic arcs increase in either case, and the reverse orientation is covered by interchanging x^δ and y^γ . Thus all sixteen triangle cells for the new block and each old block pair attain two. The within-new-block digon remains tied at zero, old sectors remain sharp, and the total increment is

$$18r + 32 \binom{r}{2} = B_{r+1} - B_r.$$

Consequently $|I(\tilde{h})| = B_{r+1}$, as required. $\square$

Lemma 7.4 (Exact moving-block defect preservation). *For $P = I(h)$ and $\tilde{P} = I(\tilde{h})$ as in Lemma 7.3,*

$$\delta_{A \cup \{a, b\}}(\tilde{P}) = \delta_A(P). \tag{7.3}$$

Proof. Let $\omega \in \text{Top}(P)$. Extend it by the seven bands

$$ba; \quad xa \ (x \in A); \quad ab; \quad ax \ (x \in A); \quad \omega; \quad xb \ (x \in A); \quad bx \ (x \in A), \quad (7.4)$$

using any fixed old-letter order inside homogeneous bands. Directly from (7.2), every selected new arc goes forward in (7.4): arcs from xa to ab, ay , from ab to bx , from ax to old vertices or xb , from old vertices to yb , and from xb to by all respect the band order; the possible reverse-band arcs have decreasing or tied potential. The new triangle restrictions are

$$\begin{aligned} xa < ab < bx, \quad ba < ax < xb, \quad ya < ax < xy, \\ xa < ay < yx, \quad xy < yb < bx, \quad yx < xb < by. \end{aligned}$$

Thus every new digon scores one and every new triangle scores two, while old cells retain their score under ω . Taking a maximizing ω gives $\delta(\tilde{P}) \leq \delta(P)$.

Conversely, for any alphabet inclusion $A \subseteq A'$, restriction of a topological order of P' is topological for $P = P'|_A$, and new cells score at most $\beta_{|A'|} - \beta_{|A|}$. Hence

$$\max s_{A'} \leq \max s_A + \beta_{|A'|} - \beta_{|A|}, \quad \delta_{A'}(P') \geq \delta_A(P).$$

Apply this to $\tilde{P}|_A = P$ to obtain the reverse inequality in (7.3). $\square$

8 Defect-spectrum monotonicity

Let $\mathcal{D}_{m,f}$ be the set of defects attained by maximum stable predecessor DAGs for type $2^m 1^f$.

Theorem 8.1 (Support-extension monotonicity). *For every $m \geq r \geq 1$ and $f \geq 0$,*

$$\mathcal{D}_{r,0} \subseteq \mathcal{D}_{m,f}.$$

Proof. Fix $d \in \mathcal{D}_{r,0}$ and a maximum stable DAG P of defect d . Lemma 7.1 writes $P = I(h)$; Proposition 7.2 says that h is sharp. Lemma 7.3 may therefore add one moving block, and Lemma 7.4 preserves the defect exactly. Iteration gives a maximum of defect d at $(m, 0)$. Finally, iterate the canonical fixed-letter lift of Lemma 5.1 f times. It also preserves maximality and defect exactly, proving the inclusion. $\square$

Corollary 8.2 (Five persistent attained defects). *For every $m \geq 3, f \geq 0$,*

$$\{0, 1, 2, 3, 4\} \subseteq \mathcal{D}_{m,f}.$$

Proof. The exact six-letter census established below gives $\mathcal{D}_{3,0} = \{0, 1, 2, 3, 4\}$. Apply Theorem 8.1 with $r = 3$. Only the five existence statements are used here; the exhaustive upper-bound part of the census is not transported. $\square$

Remark 8.3 (Logical boundary). The theorem does not say that $\mathcal{D}_{m,f}$ equals the five displayed values, that its maximum is four, or that it is an interval. It transports neither realization-cone facets nor adjacency, connectivity, extrema reachability, or Lipschitz properties.

9 Five explicit finite seeds and the computational boundary

At $q = 6, \sigma = (01)(23)(45)$, the 150 Kautz words form 75 two-element ρ -orbits in lexicographic order. In this finite census, a *labelled maximum mask* is the 75-bit incidence vector of a maximum ρ -stable acyclic predecessor-arc set; it has weight 43 and therefore selects 86 arcs. A *source triple* is the ordered triple of local four-letter pair-core states on block pairs

(01), (02), (12). Each local state lies in a canonically ordered 36-element list, so indices (i_{01}, i_{02}, i_{12}) have radix-36 code $36^2 i_{01} + 36 i_{02} + i_{12}$. A *closure-centralizer work unit* is an equivalence class of labelled maximum masks whose transitive closures agree after relabelling by the full centralizer of σ . A *mask key* is the zero-padded 19-digit hexadecimal encoding of the 75-bit vector. In the last table column, *relations/covers* means the cardinalities of the transitive closure and transitive reduction, respectively. The following deterministic certificates give one witness for each defect.

defect	score	mask key	source triple	relations/covers
0	95	06d87f4fef6017ea3d6	5961	334/40
1	94	25d86fc9dfe7f1027e4	25902	332/42
2	93	06dc074fcf67d3ea3f4	26734	350/38
3	92	24dc67cbdfecf3823e2	12907	334/56
4	91	25d82fcdcffff2007f0	38610	362/48

The associated optimal predecessor orders and their bad cells are:

$d = 0$: 01 03 04 31 14 41 10 12 13 15 53 34 42 23 43 30 32 25
35 45 52 20 24 54 40 05 50 02 21 51

No bad cell.

$d = 1$: 01 04 31 42 52 21 14 24 41 10 12 13 15 23 25 45 54 40
43 30 32 20 05 50 02 03 53 34 35 51

Bad cell T_{235}^- .

$d = 2$: 01 31 12 52 24 41 42 21 15 23 25 51 10 13 14 32 20 45
50 04 54 40 03 05 43 35 53 30 02 34

Bad cells T_{234}^-, T_{235}^- .

$d = 3$: 12 35 40 02 52 20 01 05 21 23 25 51 10 14 45 50 03 04
53 32 24 54 41 13 43 30 31 15 34 42

Bad cells $T_{015}^+, T_{123}^-, T_{245}^+$.

$d = 4$: 12 50 02 20 24 40 01 03 04 32 34 41 10 05 42 23 25 45
54 43 35 51 52 21 13 15 53 30 31 14

Bad cells $T_{012}^-, T_{013}^-, T_{145}^-, T_{234}^-$.

Each order respects the complete canonical precedence relation encoded by its mask, and Lemma 3.2 gives the displayed defect. The complete transitive closures, reductions, centralizer transports, opposed dynamic-programming certificates, and hashes are stored in a machine-readable certificate bundle.

The existence list alone does not prove that no larger defect occurs. The population is reconstructed independently of the stored labels and mask table from the combinatorial definition, using the frozen quotient-native Route B kernel. This is a label-independent replay, not a second implementation of that kernel. The portable enumerator first derives the 36 local four-letter states, forms all $36^3 = 46,656$ source triples, quotients them into 987 centralizer orbits, and visits exactly 54,078 nodes in the complete leaf-canonical triangle search. It obtains 1,759 canonical representative masks and expands them under the 48-element centralizer to exactly 83,736 distinct labelled maximum masks. Only after this enumeration does it open the tracked table and verify exact set equality and digest.

The separate portable exhaustive optimization then covers all 83,736 labelled maximum masks and all 1,759 closure-centralizer work units. For every work unit it checks opposed

prefix and suffix dynamic-programming tables state by state against their Bellman recurrences, verifies that both routes give the same optimum, and checks a concrete optimal topological order. Its labelled-mask multiplicities for defects $0, \dots, 4$ are

$$(20976, 26640, 19896, 14208, 2016),$$

and its work-unit multiplicities are

$$(445, 555, 420, 297, 42).$$

That computation proves the upper boundary only at $(m, f) = (3, 0)$. The five certificates make the existence witnesses portable; they are not a replacement for the exhaustive census.

10 Exact obstruction clutters and lower certificates

For distinct letters $a < b < c$, let T_{abc}^+ and T_{abc}^- denote the two oriented triangle cells, with cyclic comparisons

$$\begin{aligned} T_{abc}^+ : & \quad ab < bc, \quad bc < ca, \quad ca < ab, \\ T_{abc}^- : & \quad ac < cb, \quad cb < ba, \quad ba < ac. \end{aligned}$$

A total order makes either one or two comparisons in each cell true; the cell is bad in the former case. Let $\mathcal{T}_q$ be the set of all these oriented triangle cells. In this section formula (3.1) defines $\delta_H(P)$ for any acyclic predecessor arc set, not necessarily a maximum one. For such a set P and $S \subseteq \mathcal{T}_q$, write $\text{Good}_P(S)$ when some order in $\text{Top}(P)$ makes every cell of S good. Define the obstruction clutter

$$\mathcal{H}(P) = \min_{\subseteq} \{S \subseteq \mathcal{T}_q : \text{Good}_P(S) \text{ fails}\}, \quad (10.1)$$

where only inclusion-minimal members are retained. These are cyclic-order constraints in the sense of [3, 7]; their incidence vectors lie naturally in the linear-ordering-polytope setting of [5]. The following identity is elementary and does not import a result from those papers.

Theorem 10.1 (Exact obstruction-clutter formula). *For every acyclic predecessor arc set P ,*

$$\delta_H(P) = \tau(\mathcal{H}(P)), \quad (10.2)$$

where the right-hand side is the transversal number of the clutter.

Proof. For $\omega \in \text{Top}(P)$, let $B(\omega)$ be its set of bad oriented triangle cells. This set meets every member of $\mathcal{H}(P)$, since a disjoint obstruction would be good under ω . Lemma 3.2 therefore gives $\delta_H(P) \geq \tau(\mathcal{H}(P))$.

Conversely, let B be a minimum transversal. The set $\mathcal{T}_q \setminus B$ contains no member of $\mathcal{H}(P)$. If it were infeasible, finiteness would give an inclusion-minimal infeasible subset, contradicting that assertion. Hence some $\omega \in \text{Top}(P)$ makes every cell outside B good, so $B(\omega) \subseteq B$. The bad-triangle formula yields the reverse inequality. $\square$

Let

$$F(P) = \{T \in \mathcal{T}_q : \text{Good}_P(\{T\}) \text{ fails}\}.$$

On $\mathcal{T}_q \setminus F(P)$, form the incompatibility graph $G(P)$ by joining T and U when $\text{Good}_P(\{T, U\})$ fails.

Corollary 10.2 (Forced cells plus matching). *For every acyclic P ,*

$$\delta_H(P) \geq |F(P)| + \nu(G(P)). \quad (10.3)$$

Proof. The singleton members of $\mathcal{H}(P)$ are precisely the forced cells. After deleting them, every edge of $G(P)$ is a two-element member of the clutter. A matching gives pairwise disjoint such obstructions, so every transversal contains all forced cells and at least one endpoint of each matching edge. Apply Theorem 10.1. $\square$

The lower certificate (10.3) is not always exact. The following small control prevents higher-rank obstructions from being silently discarded.

Proposition 10.3 (A rank-three obstruction). *On four letters, let P be the partial order generated by*

$$\begin{aligned} 01 < 13, \quad 03 < 20, \quad 03 < 32, \quad 10 < 31, \quad 12 < 01, \\ 20 < 23, \quad 21 < 32, \quad 31 < 20, \quad 32 < 02, \quad 32 < 30. \end{aligned} \tag{10.4}$$

Then $F(P) = \emptyset$, $G(P)$ has no edge, but $\delta_H(P) = 1$. Its unique minimal obstruction is

$$\{T_{012}^+, T_{023}^-, T_{123}^-\}.$$

Thus equality in (10.3) is false in general.

Proof. Direct enumeration gives 99,000 linear extensions of P , 162 with exactly one bad cell, and 176 distinct bad-cell sets. It verifies that every singleton and every pair of cells is simultaneously good, while every proper subset of the displayed triple is feasible. The triple itself has a short certificate: if all three cells were good, the base relations leave only compatible cyclic chains containing respectively

$$20 < 01, \quad 32 < 20, \quad 13 < 32.$$

Together with $01 < 13$, these give $20 < 01 < 13 < 32 < 20$, a contradiction. Hence the displayed triple is the unique minimal obstruction and Theorem 10.1 gives $\delta_H(P) = 1$. $\square$

11 A six-letter four-defect obstruction

We now apply the forced-cell-plus-matching certificate to a six-letter core.

Lemma 11.1 (Ten-relation four-defect obstruction). *Every total order containing the precedence relations*

$$\begin{aligned} 02 < 10 < 21, \quad 03 < 10 < 31, \\ 40 < 54 < 15, \quad 41 < 05, \\ 24 < 32 < 53, \quad 25 < 43 \end{aligned} \tag{11.1}$$

has at least four bad cells among

$$T_{012}^-, T_{013}^-, T_{045}^-, T_{145}^-, T_{234}^-, T_{235}^-.$$

Proof. The chain $02 < 10 < 21$ gives the truth pattern $(1, 0, 0)$ in T_{012}^- , so this cell is bad. Likewise $03 < 10 < 31$ makes T_{013}^- bad.

Suppose that both T_{045}^- and T_{145}^- were good. Since $40 < 54$, goodness of the first cell would force $40 < 05 < 54$. Since $54 < 15$, goodness of the second would force $54 < 41 < 15$. Hence $05 < 54 < 41$, contrary to $41 < 05$. At least one of these two cells is therefore bad.

Similarly, goodness of T_{234}^- would force $24 < 43 < 32$, while goodness of T_{235}^- would force $32 < 25 < 53$. Together they imply $43 < 32 < 25$, contrary to $25 < 43$. At least one of the last two cells is bad. The three disjoint contributions are consequently $2 + 1 + 1 = 4$. Equivalently, if P is the partial order generated by (11.1), the first two cells lie in $F(P)$, while $\{T_{045}^-, T_{145}^-\}$ and $\{T_{234}^-, T_{235}^-\}$ are disjoint edges of $G(P)$; Corollary 10.2 gives the same lower bound. $\square$

The argument is invariant under relabelling and remains valid after adding vertices or further precedence relations, provided the displayed core is retained. It does not assert that the core occurs for every involution or in every extremal order cone.

Proposition 11.2 (Exact occurrence in the six-letter census). *Let $\sigma = (01)(23)(45)$. Among the 1,759 closure-centralizer classes in the six-letter census, exactly 24 contain an image of the core (11.1) under $C_{S_6}(\sigma)$. Each of these 24 classes has defect four, each contains a unique centralizer image of the core, and none of the remaining 1,735 classes contains such an image. The 24 classes account for 1,152 labelled masks.*

Computer verification. A separate checker reconstructs the 48-element centralizer, decodes every closure, tests all centralizer images of (11.1), and independently checks all 1,152 labelled members against their canonical representatives. It records 24 unique hits, no hit outside the selected classes, and no failed transport. The commands and certificate boundary are given in Appendix A. $\square$

Remark 11.3 (Minimality boundary). Within this fixed six-cell decomposition, deleting any one of the ten relations lowers the forced minimum in its component, so (11.1) is inclusion-minimal for the displayed $2 + 1 + 1$ proof. It is not claimed to be a globally smallest classifier of the 24 finite classes.

12 Finite diagnostics and open mechanism questions

We first fix the geometric terminology used in this section. For a fixed involution σ , let

$$V_{\sigma}^{-} = \{h : A_q \rightarrow \mathbb{R} : h(jv) = -h(v)\}, \quad \ell_{xyz}(h) = h(yz) - h(xy).$$

The forms are constant on the ρ -orbits of words. After removing the structural zero forms and identifying proportional forms, their kernels are the *primitive walls*. A source mask M is *sharp* when it selects the maximum possible number of positive nonzero ρ -orbits; its relative open realization cone is

$$K(M) = \{h \in V_{\sigma}^{-} : \ell_o(h) > 0 \ (o \in M), \quad \ell_o(h) < 0 \ (o \notin M \cup Z)\},$$

where Z is the structural-zero set. A *chamber* is the positive projectivization $K(M)/\mathbb{R}_{>0}$ of a nonempty sharp cone.

Two labelled chambers are *raw sharp facet neighbours* when the topological closures of their cones share a relative codimension-one face. The certificate used here supplies exact interior potentials on both sides whose joining segment meets exactly one primitive wall and no other wall. The full centralizer $G = C_{S_q}(\sigma)$ relabels masks and cones; its vertex quotient has the chamber orbits GM , while raw edges retain their labelled endpoints and crossed wall. At $q = 6$, $G \cong C_2 \wr S_3$ has order 48. A *candidate flip* switches the complete sign class of one of the 18 primitive walls having one form in each orientation. It becomes a realized neighbour only when the flipped sharp mask has a certified nonempty cone and the exact one-wall segment test passes.

A bounded diagnostic does not enter the preceding proofs. At $m = 3$, two certified sharp facet adjacencies have endpoint defects $(0, 3)$ and $(4, 1)$. Thus defect is not 1-Lipschitz, and the maximum facet jump is at least three. A bounded panel of 90 candidate flips finds 23 realized neighbours, with jump histogram

$$0 : 13, \quad 1 : 4, \quad 2 : 4, \quad 3 : 2.$$

This panel is not the global facet graph and does not determine connectivity or the maximum jump. The obstruction of Section 11 explains 24 of the 42 defect-four symmetry classes at $(3, 0)$; it does not classify the other 18 or provide an all-parameter defect formula.

A Reproducibility and artifact boundary

The finite seed certificates are reproduced and independently checked by

```
python -B companion/verify_five_defect_witnesses.py
python -B -O companion/verify_five_defect_witnesses.py
```

These five certificates prove attainment. The reverse inclusion $\mathcal{D}_{3,0} \subseteq \{0, 1, 2, 3, 4\}$ comes instead from the complete six-letter verification described in Section 9. Population completeness is first reconstructed from the 36 local states and all 46,656 source triples by

```
python -B companion/verify_population.py
python -B -O companion/verify_population.py
```

The checker enumerates 987 source orbits and 1,759 canonical representative masks, expands them to 83,736 labelled masks, and consults the tracked mask rows only for a final exact-set comparison. It imports neither Route A nor a stored population as an enumeration oracle; it deliberately reuses the frozen quotient-native Route B kernel rather than claiming a second implementation.

The portable bundle contains opposed dynamic-programming tables for all 1,759 closure-centralizer classes. The following standard-library checker reconstructs one authenticated canonical closure per class, expands the 1,759 representative mask orbits under all 48 centralizer actions, verifies the mask-to-class assignment of every one of the 83,736 tracked masks, checks all 3,518 tables against their Bellman recurrences, checks route agreement and explicit optimal orders, and derives the displayed spectrum:

```
python -B companion/verify_bellman_bundle.py
python -B -O companion/verify_bellman_bundle.py
```

The public replay uses no historical SQLite database.

The four-letter rank-three obstruction of Proposition 10.3 is reproduced by

```
python -B companion/verify_rank_three_obstruction.py
python -B -O companion/verify_rank_three_obstruction.py
```

The standard-library checker enumerates all 99,000 linear extensions, reconstructs the 176 attained bad-cell sets, verifies that forced cells and pairwise incompatibilities are absent, and recovers the displayed triple as the unique inclusion-minimal obstruction.

The exact occurrence statement in Proposition 11.2 is reproduced by

```
python -B companion/verify_ten_relation_core.py
python -B -O companion/verify_ten_relation_core.py
```

The checker reconstructs each canonical closure directly from the tracked 75-bit source masks and tests the core on all 1,759 work identities. It discovers 24 classes and all their 1,152 labelled members. The human $2 + 1 + 1 = 4$ lower bound and two directly checked score-91 topological orders per discovered class give matching lower and upper bounds.

The bounded facet diagnostic in Section 12 is reproduced by

```
python -B companion/verify_facet_counterexample.py
python -B -O companion/verify_facet_counterexample.py
```

These commands certify the two displayed sharp adjacencies and the stated 90-flip panel only. They do not construct the full six-letter facet graph. The exact file manifest, prerequisites, expected outputs, PDF build, and clean-clone procedure are given in REPRODUCE.md at repository root.

References

- [1] Oleksiy Babansky. A parity theorem for reverse-complement-invariant minimum decycling sets in q -ary de bruijn graphs. Manuscript and reproducibility package, version 2.0.0, 2026. Accessed 2 September 2026.
- [2] K.-S. Cheung and Ernest S. Kuh. Feedback vertex and edge sets of a digraph and applications to sparse matrices. Technical Report UCB/ERL M419, Electronics Research Laboratory, University of California, Berkeley, 1974. Section III, Theorem 2, pp. 5–6.
- [3] Zvi Galil and Nimrod Megiddo. Cyclic ordering is NP-complete. *Theoretical Computer Science*, 5(2):179–182, 1977.
- [4] Andrew V. Goldberg and Alexander V. Karzanov. Maximum skew-symmetric flows and matchings. *Mathematical Programming*, 100(3):537–568, 2004.
- [5] Martin Grötschel, Michael Jünger, and Gerhard Reinelt. Facets of the linear ordering polytope. *Mathematical Programming*, 33(1):43–60, 1985.
- [6] Guillaume Marçais, C. S. Elder, and Carl Kingsford. k -nonical space: Sketching with reverse complements. *Bioinformatics*, 40(11):btae629, 2024.
- [7] J. Opatrny. Total ordering problem. *SIAM Journal on Computing*, 8(1):111–114, 1979.
- [8] Jun-Ming Xu, Ye-Zhou Wu, Jia Huang, and Chao Yang. Feedback numbers of Kautz digraphs. *Discrete Mathematics*, 307(13):1589–1599, 2007.
- [9] Xirong Xu, Chun Yin, Sijia Zhang, and Yazhen Huang. Improved feedback vertex sets in Kautz digraphs $K(d,n)$. In *2014 Tenth International Conference on Computational Intelligence and Security*, pages 161–165, 2014. IEEE document 7016874.
- [10] Chun Yin. Research on feedback numbers of Kautz graph $K(d,n)$ and Knoedel graph $W(4,n)$. Master’s thesis, Dalian University of Technology, 2015. Catalogue and abstract consulted; full text unavailable.
- [11] David H. Younger. Minimum feedback arc sets for a directed graph. *IEEE Transactions on Circuit Theory*, 10(2):238–245, 1963.
- [12] Sijia Zhang, Xirong Xu, Yuansheng Yang, and Chun Yin. Improved algorithm for feedback vertex number in Kautz digraphs $K(d,n)$. *Journal of Chinese Computer Systems*, 37(10):2273–2278, 2016. Page range cross-checked against the issue table of contents; full text unavailable.